Joshua Santiago

Construction Coordinator / AI Developer / IT at Jacobs, Life Sciences Americas, Puerto Rico

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SUMMARY

AI-native software engineer who builds and operates multi-agent systems with Claude Code, and production software for regulated industries. The only technical person in my Jacobs office: I built the reporting platform our project runs on and am building its field app. Earlier: validated MES apps for a medical device maker, HIPAA-compliant storage for a health plan, CI/CD at Granicus.

SELECTED PROJECTS

Joshua-OS: Claude Code operating system · In production — Aug 2026 – Present

Claude Code, Markdown agents and skills, Bash, PowerShell, Python, GitHub Actions

- Wrote the SOP and 9 global rule files every project follows (plan, build, verify, ship, learn; TDD with 80% coverage, pass^k on critical paths). Built worktree.sh so each builder agent gets its own git worktree: 27 parallel lanes merged into main.
- Assembled and operate 66 Claude Code subagents (many from plugins), each with its own model and tool allow-list, plus Playwright, Exa and Supabase MCP servers; built a tested Claude Code status mod.
- 190 commits since Aug 19, 2026. Reviewer agents verify the work before a task is ticked done.

Jacobs reporting and compliance platform · In production — 2026 – Present

Python 3.11, openpyxl, FastAPI, vanilla JavaScript, hand-written SVG charts

- Built a 2,409-line Python ETL that finds each Excel and Primavera P6 workbook by its content and maps columns by header name, feeding a 26-view dashboard with 0 installs; it replaced hand-assembled reporting.
- Ported a 9,000-line React and FastAPI workforce app into the static dashboard; consolidated five compliance and commercial trackers covering over 100 workers and more than 30 credential types.
- Every number traces to a source cell; absent fields show as unavailable, never zero. Keyboard-navigable dialogs with focus return, audited against WCAG AA.

JevTrade: pre-registered LLM evaluation · Paper / sandbox — Oct 2026, five-day sprint

Python 3.12, pytest, ruff, uv, Alpaca paper API, TypeSafe Jev API

- Tested whether a commercial typed-judgment model calls BTC direction better than trivial baselines, with the pass bar fixed in advance. Scored 17,005 calls from cached replays with a 10,000-resample block bootstrap: 47.3% hit rate (95% CI 46.4–48.3) vs 50.9% for always up. The model failed, and I published it.
- Honesty harness: last 6 months sealed, walk-forward with purge, deflated Sharpe, AST tests that keep the lab out of order code. 1,275 tests passing; lab and scorer at 100% line and branch coverage, as recorded.

Jacobs field app · In progress (Stage 1 done, pre-pilot) — Sep 2026 – Present

Expo 57, React Native 0.86, TypeScript, Supabase Postgres, Jest, Playwright, GitHub Actions

- Bilingual TypeScript app for iOS, Android and web (9 screens): Spanish and English ship with 173 keys each; CI fails the build if they drift.
- Supabase Postgres: 12 tables with row-level security on every one, 25 policies, MFA enforced at the database (aal2). An importer CLI with a fidelity oracle recounts app data against the source: 23 of 23 checks matched.
- CI on GitHub Actions: typecheck, lint, Jest, pgTAP, Playwright with axe, gitleaks, actions pinned to SHAs. 461 Jest, 316 database and 21 end-to-end tests, as recorded.

EXPERIENCE

Construction Coordinator / AI Developer / IT — Apr 2026 – Present

Jacobs, Life Sciences Americas, Construction Services · Guaynabo, Puerto Rico

- Designed and built the project's reporting and compliance platform (above), which the project team adopted as its working reporting; I continue to extend it as operational needs evolve.
- The only technical person in the office: I build its dashboards, reports, workflows and field app, and fix IT.

- Coordinate construction delivery and HSE compliance for a large capital program (certifications, inductions, job hazard analyses, prequalification); gather requirements from leadership, project controls and HSE.

Application Developer and Validation Engineer Dec 2025 – Mar 2026

Validation Resource Group · medical device manufacturing client · Cayey, Puerto Rico

- Designed, developed and validated Tulip MES apps replacing paper shop-floor processes, inspection logs and electronic batch record workflows.
- Converted SOPs and validation protocols into step-driven apps with batch tracking, equipment data capture, input validation, conditional logic and audit trail capture.
- Supported system testing, User Acceptance Testing and change tracking in Jira ahead of production release.

Back End Developer I Jun 2025 – Aug 2025

Alpine Health Technology · Guaynabo, Puerto Rico

- Configured secure data storage and backups to meet HIPAA requirements for a regional health plan client.
- Optimized SQL queries across health plan applications, maintained full stack apps, and supported Azure DevOps CI/CD.

Software Engineer May 2024 – Dec 2024

Granicus · San Juan, Puerto Rico

- Implemented CI/CD pipelines in Azure DevOps and GitHub Actions to automate testing, integration and deployment of ASP.NET applications.
- Built and maintained full stack web apps across .NET and Java back ends for public sector education and justice case management systems (C#, VB.NET, Java, JavaScript, SQL Server).
- Wrote technical, user and API documentation; shipped features with design and product teams.

Software Engineer Intern Oct 2023 – Apr 2024

Granicus · San Juan, Puerto Rico

- Improved CI/CD deployment quality through code review and automated testing enhancements; supported full stack maintenance in C#, SQL and JavaScript.

SKILLS

Agents & tooling: Claude Code (primary development environment), subagents and skills, MCP servers (Playwright, Exa, Supabase), tool allow-lists, LLM evaluation (pre-registration, bootstrap CIs, cached replays), Exa, Firecrawl

Languages & frameworks: Python, TypeScript, JavaScript, Node.js, C# and .NET, SQL, Java (read and debug), C, FastAPI, pydantic, React, React Native and Expo, Django, HTML and CSS

Data & infra: PostgreSQL and Supabase (row-level security), Microsoft SQL Server, SQLite, ETL with openpyxl, Azure (AZ-204), AWS, Docker, Git, GitHub Actions, Azure DevOps, Linux

Practices: test-driven development (pytest, Jest, pgTAP, Playwright with axe), CI/CD with secret scanning (gitleaks), MES and electronic batch records, validation protocols, audit trails, UAT, HIPAA data handling, Agile, Jira

EDUCATION

Bachelor of Science, triple major in Computer Information Systems, Accounting, and Business Administration 2017 – 2022

Peru State College

Software Engineering, 9 Month Certificate Program Jun 2023 – Mar 2024

Holberton Coding School, San Juan, Puerto Rico · Capstone: HealHub, a health app (Node.js, Python, Django, PostgreSQL, AWS)

CERTIFICATIONS

- Microsoft Certified: Azure Developer Associate (AZ-204), June 2024
- Tulip Advanced Developer Certification, Dec 2025 · Tulip Basic Developer Certification, Dec 2025
- Cybersecurity Certification (Holberton), Mar 2024

LANGUAGES

Spanish: native · English: fluent